Meningioma Progression: A Meta Analysis Of Expression Profiles Comparing Grade I and Grade III Tumours

By Conor Murphy

Supervised by Dr Vasileios Panagiotis Lenis

Word Count: 4373

Abstract

Meningiomas are the most common CNS tumors, yet molecular drivers of progression from benign (grade 1) state to malignant (grade 3) state are poorly understood. This study used a meta-analysis of microarray and validated its results using RNA-sequencing data. Key pathways and genes contributing to meningioma progression were described.

Three datasets were used: GSE74385, GSE16581, GSE136661. These were acquired from the Gene Expression Omnibus. Differential gene expression was carried out using Limma for microarray data and DESeq2 for RNA-Sequencing data. The meta analysis used Fishers combined P value methods. Differentially expressed genes were defined at a cut off of Padj < 0.05 and having a log fold change >1. Gene ontology over-representation enrichment analysis was performed using WebGestalt packages and uncovered biological processes and functions linked to progression of meningioma.

950 differentially expressed genes, including AURKA, NEK2, TOP2A, were found and linked to chromosome instability and spindle organisation. AURKB and BUB1B were also found to be implicated in impaired mitotic control. Angiogenesis and related GO terms such as vascular control driven by genes SLIT2 and EDN3 suggest vascular restructuring in meningiomas drives malignancy.

This study revealed critical pathways for meningioma progression and malignancy, of particular novelty those involved in chromosome instability and vascular proliferation. These findings offer promising targets for therapeutics, biomarkers, and further studies investigating meningioma progression.

Introduction

Meningiomas are the most common central nervous system (CNS) tumour, and account for up to 37.6% of all neoplasms of the CNS [1]. These are rare in paediatric patients and mainly affect adults [1]. Despite its high prevalence, research into the molecular mechanisms causing the malignant progression of meningioma has been neglected due to its clinical profile, with 80% of meningiomas associated with long term survival and little neoplastic progression that mainly leaves patients unaffected, suffering little to no symptoms [1] .Most tumours of this type are discovered incidentally while performing scans for other diseases. Meningioma tumours are graded on a scale of 1-3, with 1 being considered benign , 2 known as “atypical” meningiomas that are mildly invasive and have a higher chance of recurrence after surgical intervention, and 3 known as anaplastic meningiomas that are malignant with high recurrence and mortality rates at a 50% survival rate after 5 years [2]. The transition from benign to malignant meningioma has been identified to have a significant change in gene expression [3] that leads to uncontrolled mitotic proliferation, genomic instability, and the disruption of essential cellular processes.

Current therapeutic approaches for grade 3 meningioma are often ineffective with limited treatment options [4] and cost the NHS £163m a year [5], with 34% spent on direct treatment and ~66% on appointments, social care, or end of life care. This underpins the need for the

development of new treatments and novel ways of identifying aggressive tumours earlier to prevent costs to the healthcare system and to patient lives.

Despite it's comparative neglect as a field of study, there is still a body of research attempting to identify mechanisms underlying tumour progression, with recent efforts focusing on transcriptomic profiling. Schmidt et al. demonstrated that grade meningiomas can relapse from their benign state and become aggressive [6], finding that recurrence and malignancy were associated with transcriptional profiles that were distinct from non-recurrent tumours suggesting alterations could be made to the traditional WHO meningioma grading system. Differentially expressed genes PTTG1, AURKB, and UBE2C were identified as key in the dysregulation of mitosis and cell cycle control associated with meningioma [6] and PTTG1 and LEPR were identified as independent prognostic biomarkers of progression. Altogether these suggested that targeted interventions and novel biomarkers could be used to mitigate the progression and aggression of the tumours.

Regulation of mitosis has emerged as a central theme in meningioma malignancy. The dysregulation of this process driven by kinesins such as KIFF11, KIF20A, and KIF14, has been identified in transcriptomic studies by Jungwirth et al. [7] that use microarray data and confirm with gene knockdown experiments. These kinases regulate mitotic progression ensuring accuracy in chromosomal segregation, and their overexpression has been associated with hallmarks of cancer progression such as aneuploidy and genomic instability [8]. The potential targeting of these kinases by inhibition has been shown in in vitro and in vivo studies that show 83% reduction in meningioma growth [9] demonstrating the ability of research of this type to produce potential meningioma treatments.

Additionally, studies have recently highlighted the significance of mitochondrial dysfunction in the pathogenesis of meningioma. Genes such as ST3GAL5 and ST8SIA1 [10] have been

identified as frequently dysregulated leading to altered energy metabolism and a reliance on glycolysis, a shift in cell metabolism known as the Warburg effect [11]. This effect has been extensively studied in glioblastoma and other CNS neoplasms suggesting pathways of dysfunction might be shared across tumours of the brain and that promising markers for meningioma progression may be linked to it [12].

Chromosomal alteration leading to gene expression changes has been well documented in meningioma. Deficit of chromosome 22q and subsequently the NF2 gene is a hallmark genetic event across meningioma grades [13]. Recently, Lee et al. has elucidated that the losses of chromosomes 18q, 6q, and 9q are major cytogenetic alterations that consistently occur in grade 3 meningiomas [14]. These alterations were linked to aggressive phenotypes, and higher recurrence rates. Genomic analysis of their dataset revealed five distinct molecular subtypes of meningioma that predict which tumours with recur more accurately than WHO histopathological grading and the KI-67 index that is applied to tumours broadly and not specifically in the CNS [14,15].

Angiogenetic factors have also been demonstrated to have a significant impact on meningioma progression. Vascular endothelial growth factor (VEGF) is consistently overexpressed in aggressive meningiomas [16], causing the formation of blood vessels to sustain the progression of the tumour and its growth. Inhibitions of VEGF and its associated receptors have shown efficacy in models of other brain tumours [17] such as glioma and glioblastoma. It's application to meningioma has also been explored in preclinical trials using an antiangiogenic agent to selectively inhibit VEGFR-2 specifically using apatinib [18]. Results were promising, showing a progression free survival of 85.7% [18], and once again demonstrating the importance of the identification of more biomarkers of stage 3 meningioma that could lead to the development of new treatments.

Furthermore, a study conducted by Viaene et al. has led to the identification of novel fusion transcripts that are differentially present in meningiomas [19], with grade 1 tumours that do not progress malignantly presenting more fusion transcripts than other tumours. The study, conducted by profiling the transcriptome using RNA-sequencing data and processing it via DESeq2 and gene enrichment analysis [19] identified GREM2, SNORA48 and SNORA46 as being significantly downregulated in meningioma progression. It further demonstrated that differences in immune gene expression in the tumour microenvironment correlates with the tumours WHO histopathological grade, showing that quantification of gene expression can be directly linked to tumour progression in meningioma.

Our study builds upon this body of work by performing a meta analysis of gene expression profiles from meningiomas using microarray and RNA-sequencing data, and attempting to identify molecular markers and pathways associated with progression from grade 1 to grade 3 meningiomas. The integration of multiple datasets will enhance the statistical power of this analysis and enable the uncovering of gene ontology (GO) terms and biological processes that are dysregulated in meningioma progression. We aim to investigate these crucial pathways, perform GO enrichment analysis, and validate our findings using RNA-sequencing data to identify potentially prognostic biomarkers and therapeutic targets of meningioma progression.

Materials and Methods

Dataset	Platform	Total Dataset Samples	Number of Grade I Samples	Number of Grade III Samples
GSE74385	Illumina HumanHT-12 V4.0 expression chip	62	20	28
GSE16581	Affymetrix Human Genome U133 Plus 2.0 Array	68	43	6
GSE136661	Illumina HiSeq 4000	160	121	7

Figure 1 ***Dataset Descriptions.*** *Shown are the platforms used, the total number of samples in the original study, and the amount of grade 1 and 3 samples deemed suitable for our analysis*

Data Acquisition

Three datasets, GSE74385 [6], GSE16581 [20,] and GSE136661 [21] were acquired from the Gene Expression Omnibus (GEO) repository, the largest publicly accessible gene expression database in order to investigate the gene expression profiles in meningioma. A systematic approach was applied to find these datasets, via utilisation of keywords such as "meningioma transcriptomics", "micro-array" and "RNA-sequencing" within the database of the GEO. Our criteria for inclusion required that the datasets focused on actual human meningioma samples and did not include animal models, were across the full range of meningioma progression from grade 1 – grade 3 including fully benign and malignant cases, and had sufficient sample sizes of > 5 grade 3 cases and >20 grade 1 cases to ensure significant statistical power in our analysis. Studies that did not include detailed metadata and description of data origin were discounted, as were those that included non standard expression profiling platforms that would prove problematic for downstream gene ontology analysis. All datasets acquired were

ensured to have a spread of ages between 25-90 to ensure ability to generalise to adult patients. Platforms deemed appropriate for this analysis included Illumina HiSeq 4000 and HT-12 expression chip, and Affymetrix Human Genome U133 plus [Fig 1]. This approach led to the narrowing of GSE74385 and GSE16581 were selected due to their focus on transcriptomic analysis of meningiomas across grades using high throughput microarray techniques allowing for a robust comparison of gene expression across both our grade 1 and grade 3 conditions. GGSE136661 was chosen as a validation dataset to confirm our identified pathways as it uses an RNA-sequencing approach to data generation, which allows capture of a wider range of novel transcripts due to its higher dynamic range, and its higher specificity. As shown in figure 1, samples for grade 3 tumours were lower than the grade 1 condition, this is due to the difficulty of acquiring grade 3 tumour samples from human patients due to reasons such as unwillingness to participate in studies at end of life and the relatively low incidence of grade 3 meningioma.

Differential Gene Expression Analysis

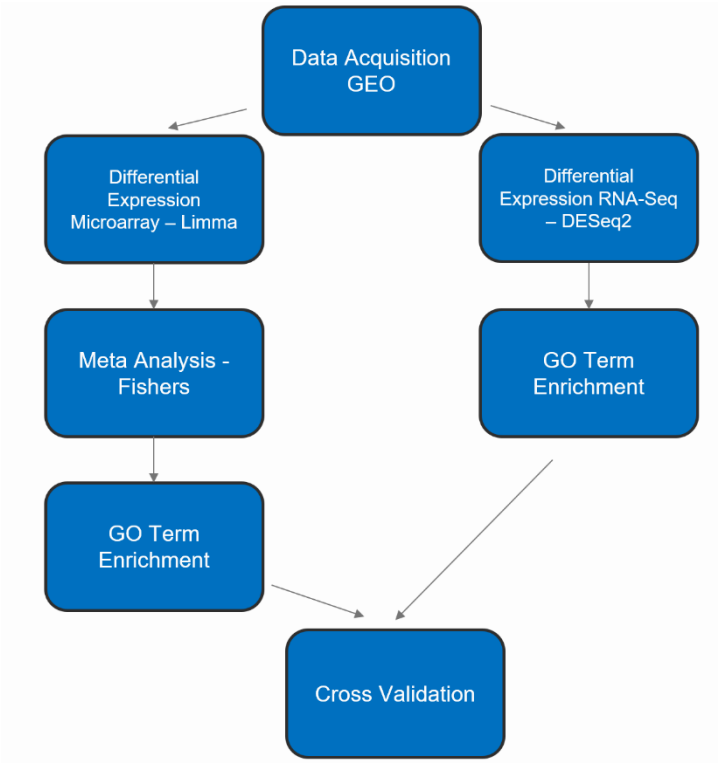


Figure 2 ***Data analysis pipeline for meningioma progression analysis****. Analysis flow for microarray data shown on the left, and flow for RNA-sequencing data shown on the right before combining.*

A comprehensive pipeline was used for analysis [Fig 2] Analysis of differential expression in genes for the microarray datasets was carried out using the Limma (Linear models for Microarray data) package in R, which is optimised for differential expression analysis using high throughput microarray datasets. Raw reads from GSE74385 and GSE16581 were pre-processed via log2 transformation. The Limma package then fits a linear model for each gene, and uses empirical Bayes methods to shrink the expected variance of gene expression values. This makes the results more stable and reliable when sample sizes are relatively small. The Benjamini Hochberg correction method for multiple testing was also applied to control for the false discovery rate and ensure reported p-values are adjusted accordingly. An adjusted P value of < 0.05 was used as a cut off for the gene being considered differentially expressed.

The RNA sequencing dataset, GSE136661 was processed using DESeq2 in R. DESeq2 models gene expression counts via the use of a negative binomial distribution. Raw counts were normalised, and differential expression of genes was found using the Wald test. Cutoffs for being significantly differentially expressed were similarly set at p-adjusted value of <0.05 and a log fold cut off of >1 was also used due to the high amount of reads generated in the RNA-seq dataset. DESeq2 was used as it incorporates shrinking of log fold change to improve the stability of our estimates, particularly for genes with low counts.

Meta Analysis

Our two microarray datasets, GSE74385 and GSE16581 were integrated using ImaGEO packages [22]. These facilitate the meta analysis of expression profiles from different platforms. It uses the Fisher's combined p-value method to integrate p values across the processed datasets, which is used to enhance the statistical power of our study. Missing data is imputed using k nearest-neighbour The process followed the standard ImaGEO protocol, using the ComBat method to remove unwanted technical variability from the different platforms used in our datasets [22]. Resulting analysis of the combined datasets identified common differentially expressed genes between the two.

Gene Ontology and Pathway Enrichment

Gene Ontology term and pathway enrichment were carried out using the WebGestalt R package. Biological processes and molecular functions were assessed. Over representation analysis was used, comparing the differentially expressed genes list obtained from the meta analysis and the DESeq2 analysis with a background data set comprising the human genome. Hypergeometric testing identified GO terms significantly enriched among these differentially expressed genes when compared to the Reactome and KEGG databases. To ensure relevance and validity of the final results of this analysis, a cutoff of a false discovery rate adjusted p

value of <0.05 was required to be considered significantly enriched. Enrichment was carried out separately for both the upregulated and downregulated genes across our two conditions.

Results

Dataset	Total Genes Analysed	Differentially Expressed Genes	Significantly Upregulated	Significantly Downregulated	Median Log Fold Change
GSE74385	28377	2174	1152	1022	0.137
GSE16581	52792	1883	1296	587	0.487
GSE136661	25280	8172	3582	4590	0.523

Figure 3 A summary statistic table of the three datasets, GSE74385, GSE16581, and GSE136661

Differential Gene Expression Analysis

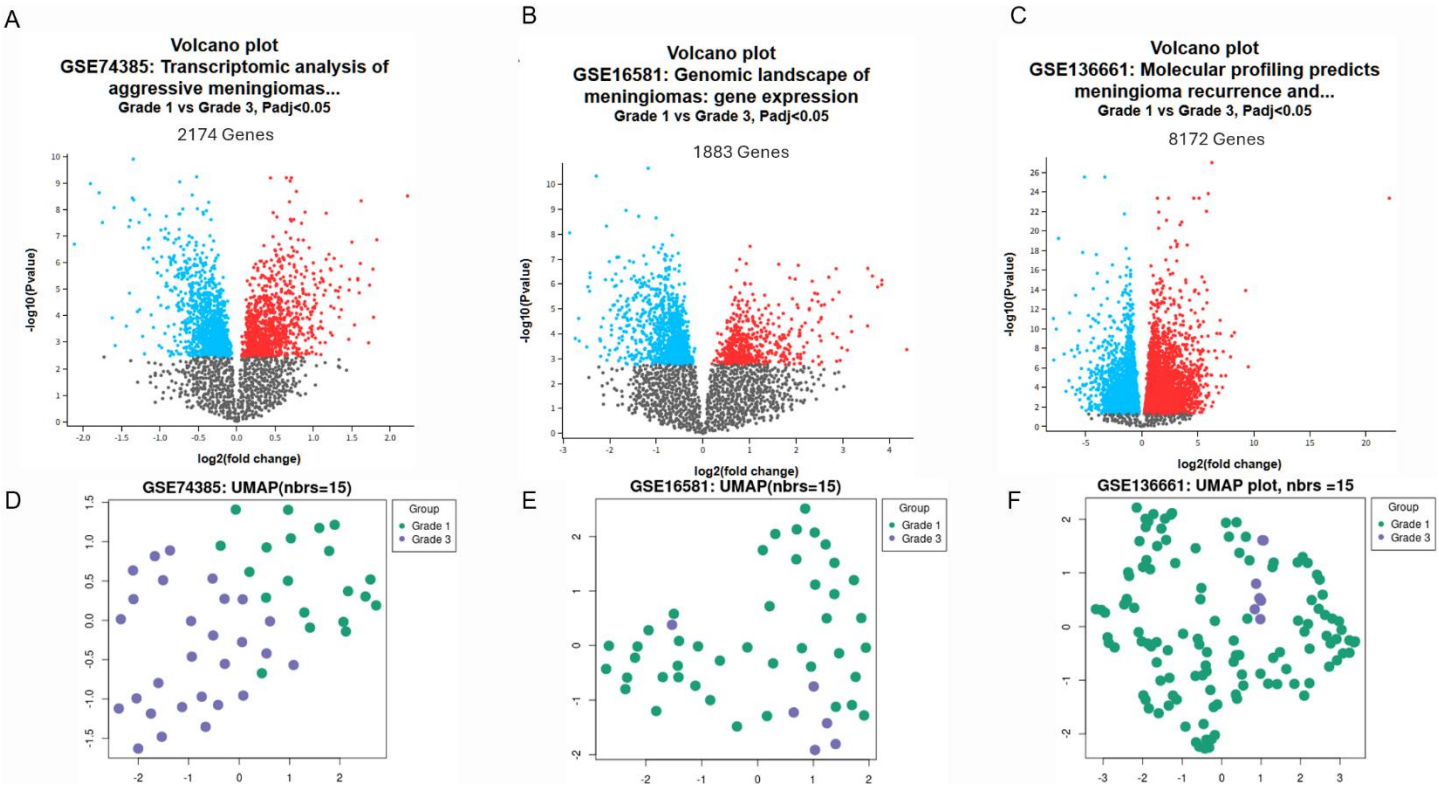


Figure 4 Initial data exploration of the differential expression analysis. A-C show Volcano plots for their respective datasets at pAdj <0.05. Upregulated genes are shown in red, and downregulated genes are shown in blue. Grey genes are those below the cutoff. D-F show UMAP plots, with grade 3 datasets shown in purple and grade 1 datasets shown in green. UMAP diagram shown to represent relations between samples after dimensionality reduction.

For the first microarray dataset, GSE74385, our analysis identified 2174 differentially expressed genes from the initial analysis of differential expression within the 48 samples. Applying a p-adj cutoff of <0.05, 2174 genes were identified as being significant [Fig 3, Fig 4a]. Visualisation of the differentially expressed genes via volcano plot revealed a significant amount of upregulation and downregulation plotting the log fold change against the -log10 P values. Among the significant genes, 1152 were upregulated and 1022 were downregulated. PCA analysis of the top genes revealed spread out but distinct clustering between grade 1 and 3 conditions, showing distinct transcriptional profiles between conditions [Fig 4D].

For our other microarray dataset, GSE16581, our analysis identified 1883 differentially expressed genes from the initial 49 samples (Padj <0.05) , of which a significant amount more, 1296, were downregulated and 587 were upregulated [Fig 3,Fig 4b]. PCA analysis showed distinct clustering between the conditions confirming the distinct profiles between our conditions [Fig 4E].

The initial DESeq2 analysis of the RNA-sequencing dataset revealed 8172 differentially regulated genes (Padj <0.05), of which 3582 were upregulated and 4590 were downregulated [Fig 3, Fig 4C]. PCA analysis revealed a wide range of genetic variety within the grade 1 datasets, whilst the grade 3 datasets clustered tightly displaying their close relations [Fig 4F].

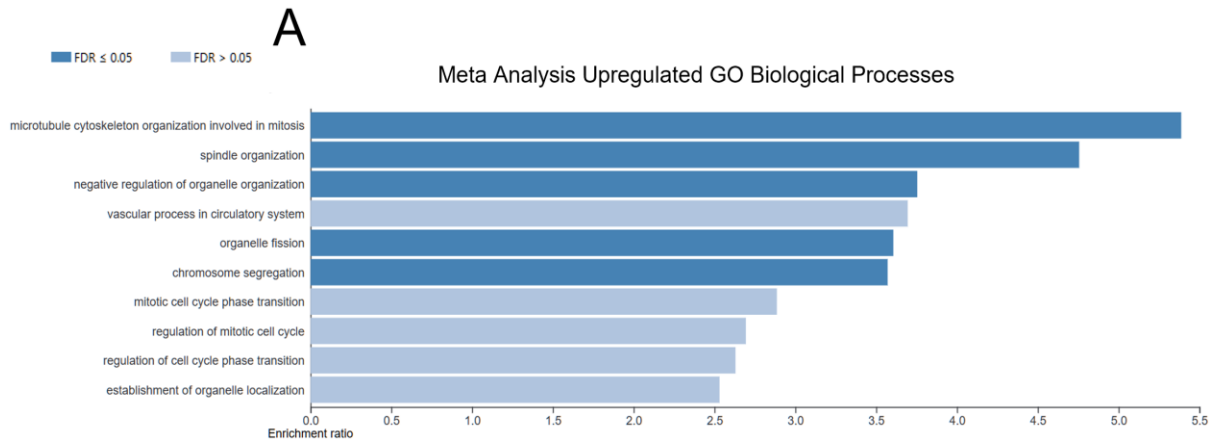
Meta Analysis

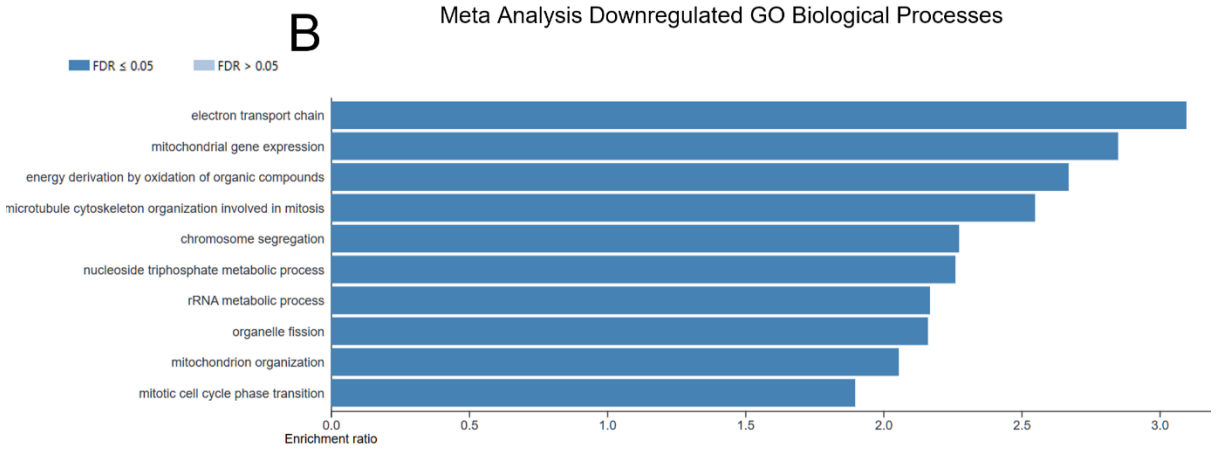
Figure 5 A Heatmap to show relation between the top 100 gene expressions in each dataset. Log2 fold count normalised to between 1, shown in dark red, to -1, shown in light green. Stage 1 samples shown in lime green at the top of the heatmap, and stage 3 samples shown in purple at the top of the heatmap.

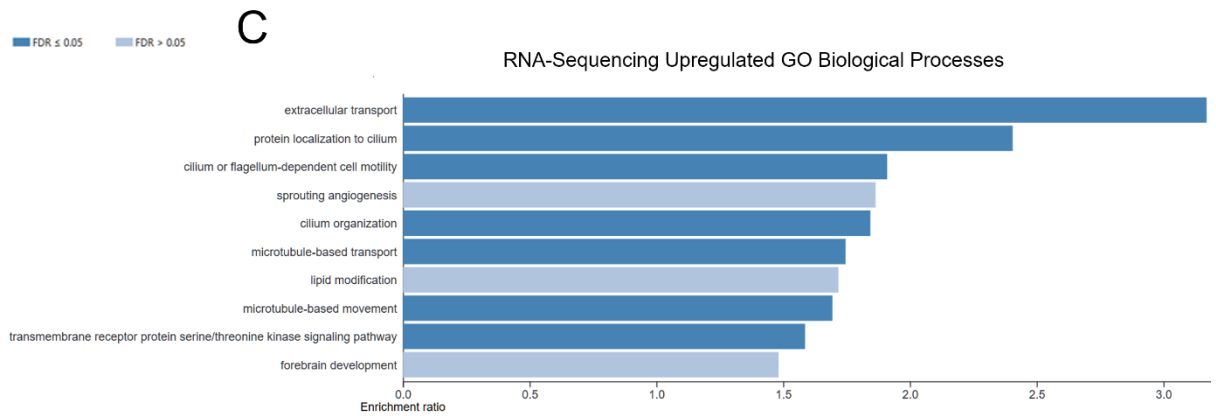
The meta analysis between GSE74385 and GSE16581 revealed 950 significant genes (Padj < 0.05), of which 260 were revealed through the combining of P-values among the datasets that would not have been deemed significantly differently expressed in the individual datasets alone. As shown in Figure 5, a distinct clustering pattern was observed on a generated heatmap of the top 100 genes found in our meta analysis. A distinction between the two

conditions of stage 1 and stage 3 meningioma was observed, represented by the clear red (upregulated) and green (downregulated) regions that are largely consistent across both datasets. The consistency in these patterns suggests that despite being generated on different patient cohorts, the datasets exhibit a very strong correlation in gene expression changes between stages of meningioma [Fig 5]. This reinforces the validity of the identified differentially expressed genes, and provides a foundation for continued exploration of the 950 identified genes are prognostic markers or targets for therapeutics in potentially malignant meningiomas. A strong cluster of downregulated genes known to be relevant to meningioma and cancer progression were identified in stage 3 tumours across the meta analysis, with genes MKI67 (-0.36 mean FC), AURKB (-0.72 mean FC), FOXM1 (-0.7 mean FC) and BUB1B (-1.1 mean FC). For upregulated genes, similar clusters of genes of interest were found such as KIFF11 (0.46 mean FC), AURKA (0.29 mean FC), TOP2A (0.82 mean FC), PITX1 (1.3 mean FC), and TK2 (1.2 mean FC).

Gene Ontology Analysis







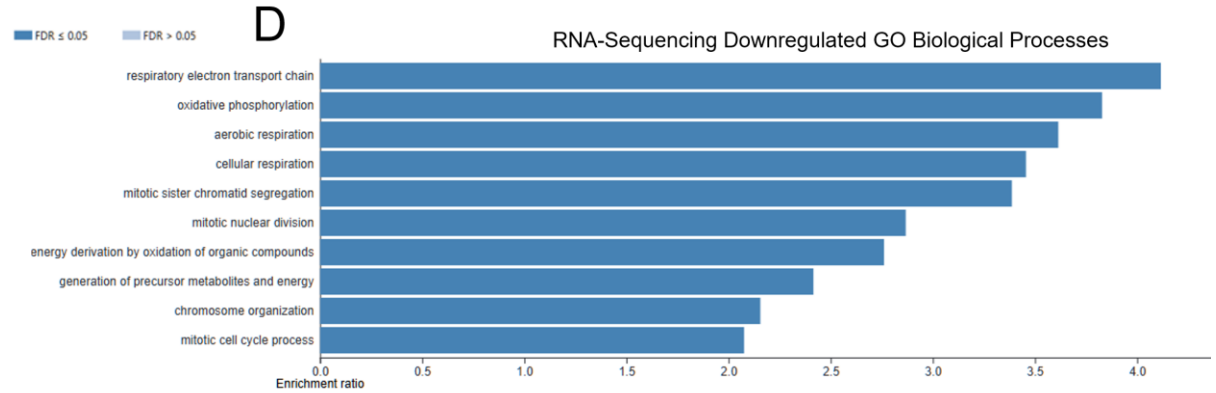


Figure 6 Gene Ontology Biological Process Terms. Conditions displayed above the chart, with FFDR < 0.05 shown in dark blue and FDR > 0.05 shown in light blue

After gene ontology enrichment, the top 10 terms from each condition – meta analysis upregulated, meta analysis downregulated, rna-seq upregulated, and rna-seq downregulated – were assessed and evaluated via false discovery rate, with FDR < 0.05 being accepted as valid and FDR > 0.05 and above terms treated with more scrutiny or rejected. Terms found in molecular function gene ontology analysis were used to support the initial conclusions drawn from biological process analysis as over 90% of the terms found by molecular function search were found to be above the threshold of 0.05 FDR.

For upregulated genes in the meta analysis condition a key finding was the upregulation of processes related to cytoskeletal organisation and spindle organisation such as spindle organisation at an enrichment ratio of 4.6, negative regulation of organelle organisation at 3.6, and microtubule cytoskeleton organization involved in mitosis at 5.4 [Fig 6A]. Key upregulated genes found to be involved with these terms include AURKA, NEK2, and TOP2A, previously mentioned genes known for aggression in meningioma. AURKA overexpression is a top value in 9 of the 10 top GO biological process terms, which is a key aspect of our findings here. The RNA-sequencing GO processes validate these findings further highlighting processes such as extracellular transport (enrichment 3.4) and protein localization to cilium (enrichment 2.4) [Fig 6C]. Alongside this, there is further validation that AURKA and NEK2 appear in many of the top processes in the RNA-sequencing analysis also displaying impact on chromosomal segregation and spindle integrity across all datasets. Further pathways of interest are those involved in angiogenesis, with vascular processes appearing in the meta analysis terms and sprouting angiogenesis in the RNA-sequencing

terms. Genes notably mentioned across both terms are SLIT2 and EDN3, validating this finding across both datasets also. These findings indicate increased angiogenesis and ciliary organisation.

For downregulated genes, the meta analysis indicated processes such as electron transport chain (3.2 enrichment), mitochondrial gene expression (2.6 enrichment), and rRNA metabolic processes (2.2 enrichment) [Fig 6B]. These pathways notably involved similar genes such as NOP58, FBL, and NOL11 which are involved in ribosome biogenesis. The RNA-Seq analysis evidently shows similar processes such as respiratory electron transport chain (4.2 enrichment), and mitotic cell cycle progression (2.1 enrichment). Genes highly involved in these processes include AURKB and BUB1B, correlating with decreased genomic stability and collapse of normal cell respiration and function.

Discussion

This study aimed to investigate the mechanisms and biological processes that underpin progression of meningioma from stage 1 to stage 3, with a focus on elucidating key genes and pathways involved in such a shift from a benign to a malignant state using transcriptomic analysis of micro-array and RNA-sequencing data. Our findings here provide significant insights into the dysfunction of normal cellular biological processes and molecular functions that drive malignancy, in particular those related to chromosomal instability, dysfunction of the cillia, and impairment of cellular metabolism, all of which have been observed to play central roles in the pathology of meningioma in the literature [23,24,25].

An important discovery made was that the upregulation of genes involved in chromosomal segregation and the organisation of spindles were consistently overrepresented across both our meta analysis and RNA-sequencing datasets. Particularly, the involvement of genes such as AURKA, NEK2, and TOP2A were high across these identified biological processes. AURKA

is largely unexplored in meningiomas [26], although it has been implicated and investigated in many other tumours such as melanoma [27] , hepatocellular carcinoma, and colorectal cancers [28]. AURKA has been implicated in some recent literature to be a marker of drug resistance [29] in both chemotherapy and in drugs used to treat other brain disorders such as Alzheimer's where AURKA is also known to be a part of regulating Amyloid B protein levels that form Alzheimer's plaques [30]. The AURKA gene is an essential mitotic kinase that in normal function ensures accurate chromosomal segregation, and its upregulation in our study in combination with recent literature [26] suggesting that it modulates ferroptosis in meningiomas make it an increasingly promising target as a biomarker of meningioma progression and as a therapeutic against said progression.

NEK2, an essential player in spindle organisation, one of our highly enriched GO biological process terms, has also been linked to instability of the genome and aneuploidy in other cancers such as lymphoma [31]. Other studies have also found it to be an essential malignant factor in glioblastoma, a lethal and aggressive brain tumour, via modulation of the NIK/NF-κB pathway [32] It's consistent appearance across multiple enriched terms in our datasets plus its apparent role in other more aggressive malignant tumours supports its potency as a biomarker for malignant progression of meningioma from grade 1 to 3, and as it is yet unexplored specifically for meningioma provides an avenue for novel research to be conducted into the efficacy of its inhibition on meningiomas as seen in studies conducted on pancreatic cancer [33]. Additionally, the link between spindle organisation terms and terms involving ciliary dysfunction in our dataset including NEK2 suggests a broad mechanism connection between processes in the dysfunction of neural cells that could lead to cancer, or has been implicated in disorders such as microcephaly [34].

Further analysis of our upregulated genes involved in identified GO pathways to do with vascular tumour involvement revealed significant involvement of the SLIT2 and EDN3 genes,

known regulators of angiogenesis [35,36]. Both genes contributed highly to the enrichment ratios of sprouting angiogenesis in the RNA-sequencing dataset and vascular circulatory system in the meta-analysis dataset. This supports the idea that grade 3 meningiomas undergo vascular restructuring and remodelling to sustain their growth and malignant patterns, especially in a subtype of meningioma known as radiation induced meningioma that is has known vascular anomalies and vascular endothelial growth factor upregulation [37] . This aligns with patterns in other brain tumours such as glioblastoma where similar vascular dysregulation is negatively correlated with patient survival [38]. These findings combined with the enriched go term of chromosomal instability suggests a dual mechanism in meningiomas driving malignancy; uncontrolled division and enhanced bloody supply to enable rapid growth of the tumour. The interlinking of these mechanisms could be a key area for future studies to explore.

Of our downregulated gene datasets, results significantly highlighted suppression of functions key to normal cell function: mitochondrial and rRNA metabolic processes. Pathways such as the highest enriched downregulated GO term electron transport chain and rRNA metabolic processes were found to be significantly affected, displaying genes such as NOP58, FBL, and NO11 as consistently downregulated across meta analysis and RNA-seq validation conditions. These genes are crucial for ribosome biogenesis and protein synthesis, which would be presumed to be impaired due to this downregulation, two concepts heavily involved in new-age cancer therapeutic studies focusing on their role in spontaneous cancer development and using it as a potential therapeutic avenue [39]. This observation in our study is further validated by the involvement of similar disruptions in Alzheimer's patients, indicating a potential link between malignant meningioma progression and broader neurodegenerative pathways [40].

Downregulation of both AURKB and BUB1B, involved as previously mentioned in chromosome segregation, was another key finding amongst our analysis. Their involvement in genomic stability and subsequent downregulation underpins the breakdown of this genetic

stability in malignant meningioma progression . These genes are well established and studies drivers of mitotic control [41,42] and their downregulation played a large part in the enrichment of GO terms in our analysis such as chromosome segregation and mitotic cell cycle. The downregulation of these markers has not been properly explored in meningioma, but has been observed as a pattern and marker of aggressive progression in glioblastoma where they accelerate proliferation and increase resistance to programmed cell death, or apoptosis [43]

Conclusion and Future Work

This study gave significant insight into biological processes and the underlying molecular genetic mechanisms driving meningioma progression from grade 1 to grade 3, or benign to malignant. These mechanisms included chromosomal instability, spindle organisation, and vascular dysregulation. Significant genes described were AURKA, NEK2, and TOP2A among others. Their vital role in these processes underpins their stated potential as therapeutic targets and biomarkers for the progression of meningioma, and is reinforced by various examples in literature of their involvement in driving progression of other neoplasms of the CNS and neurodegenerative disorders such as Alzheimer's. Furthermore, the exploration of vascular regulators SLIT2 and EDN3 driving angiogenesis in high grade meningiomas is a mechanism that also warrants further explanation, as they likely facilitate meningioma growth making them prime targets for therapeutic interventions to slow malignant progression.

To conclude, this study offers a deeper understanding of the genetic profile of meningioma progression, validated across 3 distinct datasets with differing preparation methods and populations. These can lay the foundation for future work using both in vivo and in vitro studies to fully explore the mechanisms explained which applied specifically to meningioma. These have the potential to be conducted rapidly as there is also existing literature and

evidence of the efficacy of treatment focused on these pathways in other tumours, boosting the chances of success when applied to meningioma. Further work could attempt to divide meningioma into further subtypes within grades if we can dig down and confirm specific markers that confirm progression from stage 1 to stage 3 and perhaps intervene before the tumour begins to progress. Particular attention should be given to samples from grade 1 meningiomas that resemble the molecular patterns of those from grade 3, and longitudinal studies could be conducted to see how those cases progress. Targeted studies on larger cohorts with data generated in-house would be much more applicable in its analysis too, and this work supports that transcriptomics is a valid and powerful tool for tracking genomic progression between stages in cancers.

References

1. Franca, R. A., Della Monica, R., Corvino, S., Chiariotti, L. & Del Basso De Caro, M. WHO grade and pathological markers of meningiomas: Clinical and prognostic role. *Pathology - Research and Practice* **243,** 154340 (2023).
2. Torp, S. H., Solheim, O. & Skjulsvik, A. J. The WHO 2021 classification of Central Nervous System tumours: A practical update on what neurosurgeons need to know—a minireview. *Acta Neurochirurgica* **164,** 2453–2464 (2022).
3. Maier, A. D. *et al.* Gene expression analysis during progression of malignant meningioma compared to benign meningioma. *Journal of Neurosurgery* 1–11 (2022). doi:10.3171/2022.7.jns22585

4. Walcott, B. P., Nahed, B. V., Brastianos, P. K. & Loeffler, J. S. Radiation treatment for WHO grade II and III Meningiomas. *Frontiers in Oncology* **3,** (2013).
5. Kanso, N., Le Calvez, K., Mauricaite, R. & Williams, M. P11.41.B the cost of inpatient care for adult primary brain tumour patients in England. *Neuro-Oncology* **24,** ii66–ii67 (2022).
6. Schmidt, M. *et al.* Transcriptomic analysis of aggressive meningiomas identifies PTTG1 and LEPR as prognostic biomarkers independent of WHO grade. *Oncotarget* **7,** 14551–14568 (2016).
7. Jungwirth, G. *et al.* Identification of kif11 as a novel target in meningioma. *Cancers* **11,** 545 (2019).
8. Gordon, D. J., Resio, B. & Pellman, D. Causes and consequences of aneuploidy in cancer. *Nature Reviews Genetics* **13,** 189–203 (2012).
9. Jungwirth, G. *et al.* KIF11 inhibitors Filanesib and ispinesib inhibit meningioma growth in vitro and in vivo. *Cancer Letters* **506,** 1–10 (2021).
10. Selke, P. *et al.* Glycation interferes with the expression of Sialyltransferases in Meningiomas. *Cells* **10,** 3298 (2021).
11. Zhang, X. *et al.* analysis of key genes regulating the Warburg effect in patients with gastrointestinal cancers and selective inhibition of this metabolic pathway in liver cancer cells. *OncoTargets and Therapy* **Volume 13,** 7295–7304 (2020).
12. Zhao, J. *et al.* Advancing glioblastoma treatment by targeting metabolism. *Neoplasia* **51,** 100985 (2024).
13. Ruttledge, M. H. *et al.* Deletions on chromosome 22 in sporadic meningioma. *Genes, Chromosomes and Cancer* **10,** 122–130 (1994).
14. Lee, Y. S. & Lee, Y. S. Molecular characteristics of meningiomas. *Journal of Pathology and Translational Medicine* **54,** 45–63 (2020).
15. Nahed, A. S. & Shaimaa, M. Y. Ki-67 as a prognostic marker according to breast cancer molecular subtype. *Cancer Biology & Medicine* **13,** 496 (2016).

16. Pfister, C., Pfrommer, H., Tatagiba, M. S. & Roser, F. Vascular endothelial growth factor signals through platelet-derived growth factor receptor β in meningiomas in vitro. *British Journal of Cancer* **107,** 1702–1713 (2012).
17. Stefanik, D. Vascular endothelial growth factor in malignant disease of the central nervous system. *VEGF and Cancer* 72–82 (2004). doi:10.1007/978-1-4419-9148-5_9
18. Wang, Y. *et al.* Apatinib in recurrent anaplastic meningioma: A retrospective case series and Systematic Literature Review. *Cancer Biology & Therapy* **21,** 583–589 (2020).
19. Viaene, A. N. *et al.* Transcriptome signatures associated with meningioma progression. *Acta Neuropathologica Communications* **7,** (2019).
20. Wang, J. Z., Nassiri, F., Mawrin, C. & Zadeh, G. Genomic landscape of meningiomas. *Advances in Experimental Medicine and Biology* 137–158 (2023). doi:10.1007/978-3-031-29750-2_11
21. Patel, A. J. *et al.* Molecular profiling predicts meningioma recurrence and reveals loss of dream complex repression in aggressive tumors. *Proceedings of the National Academy of Sciences* **116,** 21715–21726 (2019).
22. Toro-Domínguez, D. *et al.* ImaGEO: Integrative gene expression meta-analysis from Geo Database. *Bioinformatics* **35,** 880–882 (2018).
23. Chromosomal instability. *SpringerReference* doi:10.1007/springerreference_172797
24. Findakly, S. *et al.* Meningioma cells express primary cilia but do not transduce ciliary hedgehog signals. *Acta Neuropathologica Communications* **8,** (2020).
25. Masalha, W. *et al.* Metabolic alterations in meningioma reflect the clinical course. *BMC Cancer* **21,** (2021).
26. Ye, Y. *et al.* Meningioma achieves malignancy and Erastin-induced ferroptosis resistance through foxm1-Aurka-Nrf2 axis. *Redox Biology* **72,** 103137 (2024).
27. Zhang, X. F. & Long, S. *Aurka is a prognostic potential therapeutic target in skin cutaneous melanoma modulating the tumor microenvironment, apoptosis, and hypoxia* (2022). doi:10.21203/rs.3.rs-1528209/v1

28. Zheng, D. *et al.* Emerging roles of Aurora-a kinase in cancer therapy resistance. *Acta Pharmaceutica Sinica B* **13,** 2826–2843 (2023).
29. Miralaei, N., Majd, A., Ghaedi, K., Peymani, M. & Safaei, M. Integrated pan-cancer of aurka expression and drug sensitivity analysis reveals increased expression of aurka is responsible for drug resistance. *Cancer Medicine* **10,** 6428–6441 (2021).
30. Afghah, Z. *et al.* Involvement of endolysosomes and Aurora Kinase A in the regulation of amyloid β protein levels in neurons. *International Journal of Molecular Sciences* **25,** 6200 (2024).
31. McCrury, M. *et al. Bifunctional inhibitor reveals NEK2 as a therapeutic target and regulator of oncogenic pathways in lymphoma* (2024). doi:10.1158/1535-7163.25337945
32. Xiang, J. *et al.* NEK2 enhances malignancies of glioblastoma via NIK/NF-KB pathway. *Cell Death & Disease* **13,** (2022).
33. Zhang, X. *et al.* NEK2 inhibition triggers anti-pancreatic cancer immunity by targeting PD-L1. *Nature Communications* **12,** (2021).
34. Patwardhan, D., Mani, S., Passemard, S., Gressens, P. & El Ghouzzi, V. Stil balancing primary microcephaly and cancer. *Cell Death & Disease* **9,** (2018).
35. Li, S. *et al.* SLIT2 promotes angiogenic activity via the robo1-VEGFR2-ERK1/2 pathway in both in vivo and in vitro studies. *Investigative Opthalmology & Visual Science* **56,** 5210 (2015).
36. Wiesmann, F. *et al.* Frequent loss of endothelin-3 (EDN3) expression due to epigenetic inactivation in human breast cancer. *Breast Cancer Research* **11,** (2009).
37. Mahmoud, Q. Meningioma. Radiopaedia.org (2024). doi:10.53347/rid-183767
38. 1.Geraldo, L. H. et al. Slit2-robo signaling in tumor-associated microglia/macrophages drives glioblastoma immunosuppression and vascular dysmorphia (2021). doi:10.1101/2021.04.12.438457

39. 1.Pelletier, J., Thomas, G. & Volarević, S. Ribosome Biogenesis in cancer: New players and therapeutic avenues. Nature Reviews Cancer 18, 51–63 (2017).
40. 1.Suzuki, M. et al. Upregulation of ribosome complexes at the blood-brain barrier in alzheimer's disease patients. Journal of Cerebral Blood Flow & Metabolism 42, 2134–2150 (2022).
41. Trakala, M., Fernández-Miranda, G., Pérez de Castro, I., Heeschen, C. & Malumbres, M. Aurora B prevents delayed DNA replication and premature mitotic exit by repressing p21 cip1. *Cell Cycle* **12,** 1030–1041 (2013).
42. Davenport, J. W., Fernandes, E. R., Harris, L. D., Neale, G. A. M. & Goorha, R. The mouse mitotic checkpoint gene bub1b, a novel bub1 family member, is expressed in a cell cycle-dependent manner. Genomics 55, 113–117 (1999).
43. Alafate, W. et al. Targeting aurora kinase B attenuates chemoresistance in glioblastoma via a synergistic manner with temozolomide. Pathology - Research and Practice 215, 152617 (2019).

AI Generation Statement:

AI has been used to generate a loose plan for the structuring of the thesis, and none of its own wording or work has been submitted here in this project.